

TM Forum Technical Specifications

Service Qualification

TMFC009

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Notice

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1. Overview

Component Name	ID	Description	ODA Function Block
Service Qualification	TMFC009	Service Qualification component is responsible for checking and validating the availability of a service according to specified and configured business rules. It must identify at least one technical solution (RFSspec) available to deliver the service (CFSspec) and check the availability of all the resources types involved in this technical solution. No resources are allocated during Service Qualification. Service Qualification component has functionality that include checking service feasibility status and publishing or reporting service qualification result, but also calculated service delivery due date and identified need of an appointment at the customer site. Service Qualification can also be in charge of the cost calculation of the technical solution identified, when it cannot be determined at catalog design time (complex B2B services). This information will be used as an input to price calculation.	Production

TMFC009 : Service Qualification



2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.4.5	L2	Service Configuration & Activation	Service Configuration & Activation processes encompass allocation, implementation, configuration, activation and testing of specific services to meet customer requirements, or in response to requests from other processes to alleviate specific service capacity shortfalls, availability concerns or failure conditions. Where included in the service provider offering, these processes extend to cover customer premises equipment. Responsibilities of the Service Configuration & Activation processes include, but are not limited to: • Verifying whether specific service designs sought by customers are feasible as part of pre-order feasibility checks; • Allocating the appropriate specific service parameters to support service orders or requests from other processes; • Reserving specific service parameters (if required by the business rules) for a given period of time until the initiating customer order is confirmed, or until the reservation period expires (if applicable); • Implementing, configuring and activating specific services, as appropriate; • Testing the specific services to ensure the service is working correctly; • Recovery of specific services; • Updating of the Service Inventory Database to reflect that the specific service has been allocated, modified or recovered; • Assigning and tracking service provisioning activities; • Managing service provisioning jeopardy conditions • Reporting progress on service orders to other processes.
?	?	?	no L3 to cover standard availability/feasibility checks

Note: refer to JIRA section and the need to identify a new Service Availability Check/Assessment activity at Service level, as part of L2 Service Configuration & Activation

Note: previously identified processes at Resource level didn't exist anymore in eTOM 23.5 and no equivalent has been found. A new Jira ticket is created.

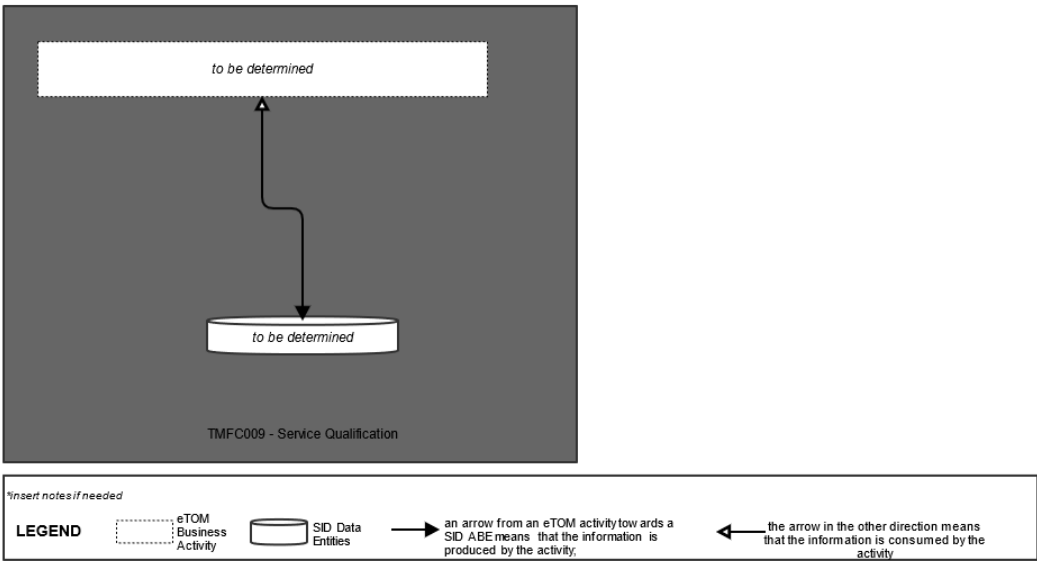
2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)

Note: refer to JIRA section and the need to create a Service Qualification BE.

2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Functions Level 1	Aggregate Functions Level 2
319	Service Feasibility Checking	Service Feasibility Checking provides checking based on the customer service location, service feasibility checks are done to assure the offering can actually be provided to the customer. This implies that the customer location is clearly established. Service feasibility checks are conducted via contract with the Service Order Management function.	Service Order Management	Service Availability
586	Service Availability Validation	Service Availability Validation function validates that the service or services specified on the service order are available at the specified customer/service location and feasible from a network point of view.	Service Order Management	Service Availability
571	Service Delivery Due Date Calculation	Service Delivery Due Date Calculation functions calculates the service delivery due date using network capacity, access provider selection and work center intelligence (including workload and capacity).	Service Order Management	<i>Service Order Initialization</i>

Note: refer to JIRA section for improvement in classification and new function required.

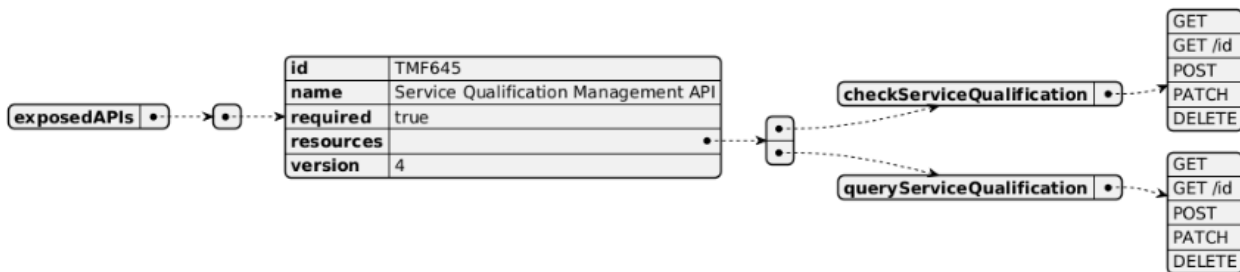
3. TM Forum Open APIs & Events

The following part covers the APIs and Events; This part is split in 3:

- List of **Exposed APIs** - This is the list of APIs available from this component.
- List of **Dependent APIs** - In order to satisfy the provided API, the component could require the usage of this set of required APIs.
- List of **Events (generated & consumed)** - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. Exposed APIs

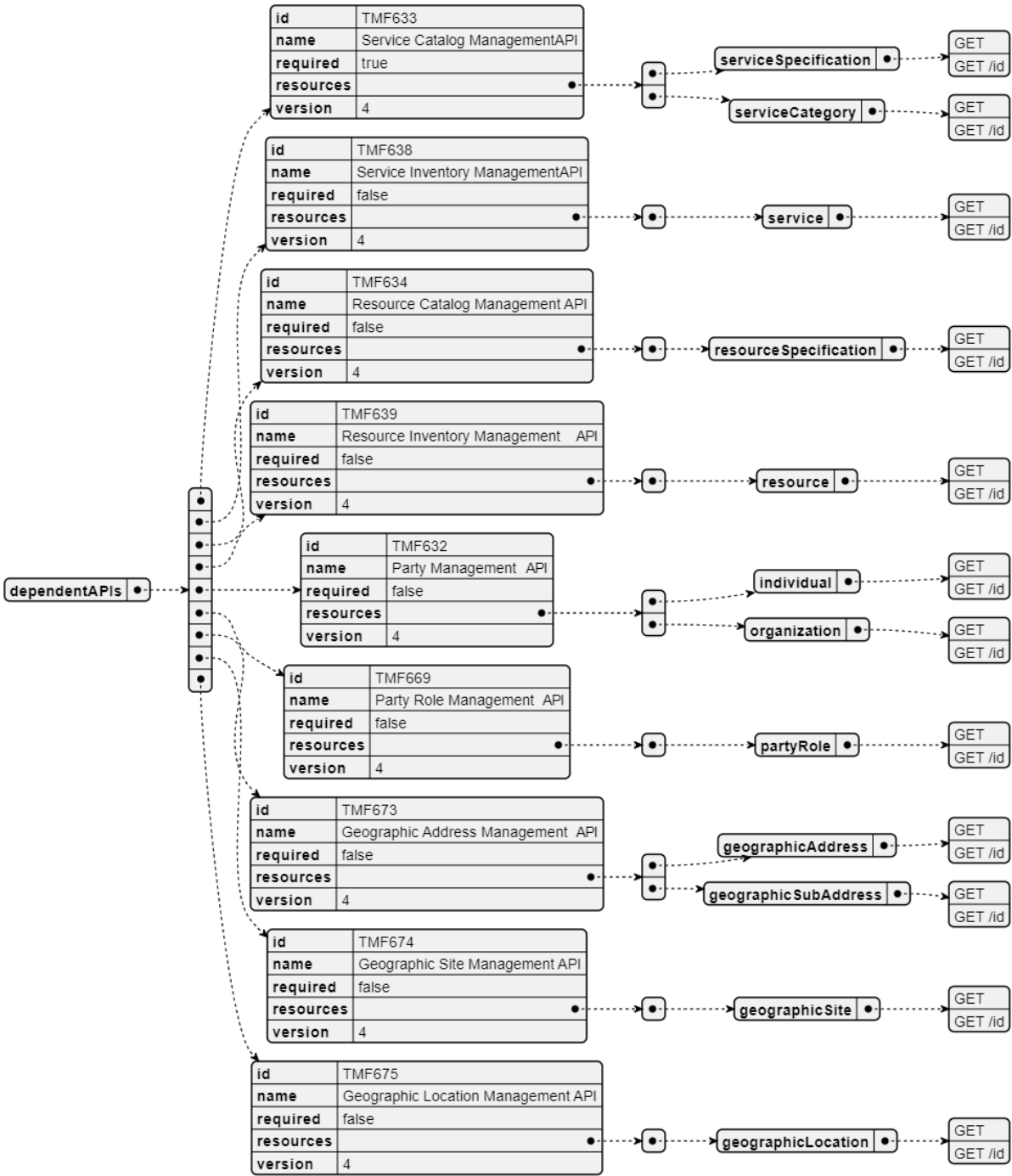
The following diagram illustrates API/Resource/Operation:



API ID	API Name	API Version	Mandatory / Optional	Operations
TMF645	TMF645 Service Qualification Management	4	Mandatory	checkServiceQualification • GET • GET /ID • POST • PATCH • DELETE
TMF645	TMF645 Service Qualification Management	4	Mandatory	queryServiceQualification • GET • GET /ID • POST • PATCH • DELETE

3.2. Dependent APIs

The following diagram illustrates API/Resource/Operation:



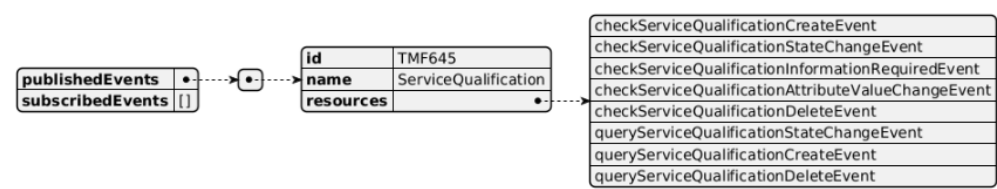
API ID	API Name	API version	Mandatory / Optional	Operations
TMF639	Resource Inventory Management	4	Optional	resource - GET - GET/id
TMF669	Party Role Management	4	Optional	partyRole - GET - GET/id
TMF632	Party	4	Optional	individual/organization - GET - GET/id
TMF672	User Roles And Permissions	4	Optional	permission - GET - GET/id
TMF673	Geographic Address Management	4	Optional	geographicAddress - GET - GET/id
TMF673	Geographic Address Management	4	Optional	geographicSubAddress - GET - GET/id
TMF674	Geographic Site Management	4	Optional	geographicSite - GET - GET/id
TMF675	Geographic Location	4	Optional	geographicLocation - GET - GET/id
TMF633	Service Catalog Management	4	Mandatory	serviceSpec - GET - GET/id
TMF633	Service Catalog Management	4	Mandatory	serviceCategory - GET - GET/id
TMF638	Service Inventory Management	4	Optional	service GET

API ID	API Name	API version	Mandatory / Optional	Operations
				GET/id
TMF688	Event Management	4	Optional	event - GET - GET/id
TMF634	Resource Catalog management	4	Optional	resourceSpecification - GET - GET/id

- GET
- GET/id

3.3. Events

The following diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.



4. Machine Readable Component Specification

Refer to the ODA Component Map on the TM Forum website for the machine-readable component specification files for this component. [TM Forum - ODA Component Directory](#)

5. References

5.1. TMF Standards related Versions

Standard	Version(s)
eTOM	23.5
SID	23.5
Functional Framework	23.5

5.2. eTOM

- identify a new Service Availability Check/Assessment activity at Service level, as part of L2 Service Configuration & Activation and able to provide all the expected results (feasibility status, service delivery due date, need of an appointment at customer site, and cost of the technical solution when needed)
- restore the Determine Resource Availability activity at Resource level. Initial definition was "This process investigates the ability to be able to satisfy specific service orders as a part of a feasibility check. Where the Allocate & Install Resource processes are requested by a pre-feasibility resource order, or by the Design Resources processes, these processes determine whether the requested resources are available."

5.3. SID

- identify a new Service Qualification BE (if we want to register the result of the Service Availability Check/Assessment)

5.4. Functional Framework

- 592 - Service Parameters Reservation should not be classified in Service Availability subdomain level 2, but rather in Service Configuration & Activation / Service Configuration
- 571 - Service Delivery Due Date calculation should be classified in Service Availability sub-domain level 2 and not in Service Order Initialization
- Add a function to calculate the cost of a technical solution, for complex cases when it cannot be done at catalog design time.

5.5. API

- TMF645 - Service Qualification API: be able to receive as an answer not only the qualification result but also the calculated due date, the need of an appointment and the cost of the solution identified.

5.6. Further resources

As listed in <https://www.tmforum.org/resources/standard/ig1214-mapping-of-oda-components-and-oda-use-cases-v1-0-0/>, this component is involved in the following use cases described in <https://www.tmforum.org/resources/standard/ig1228-how-to-use-oda-using-open-apis-to-realize-use-cases-v9-0-0/> :

Refer to: UC002, UC003, UC007, UC008 and UC010

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
1.0.0	05-Aug-2022	Goutham Babu	Final edits prior to publication
1.0.1	25 Jul 2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.1.0	13 May 2024	Gaetano Biancardi	Component Template Update odf 23.5 update TMF688 removed from exposed and dependent API
1.1.0	12 Jul 2024	Amaia White	Final updates before publication

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	05-Aug-2022	Goutham Babu	Initial release of document
Pre-production	05-Sep-2022	Adrienne Walcott	Updated to reflect TM Forum Member Evaluated status
Pre-production	14-Aug-2023	Amaia white	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-Production	12-Jul-2024	Amaia White	New release 1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgments

Team Member (@mention)	Company	Role*
Sylvie Demarest	Orange	Editor
Ian Turkington	TM Forum	Additional Inputs
Hugo Vaughan (TM Forum)	TM Forum	Additional Inputs

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